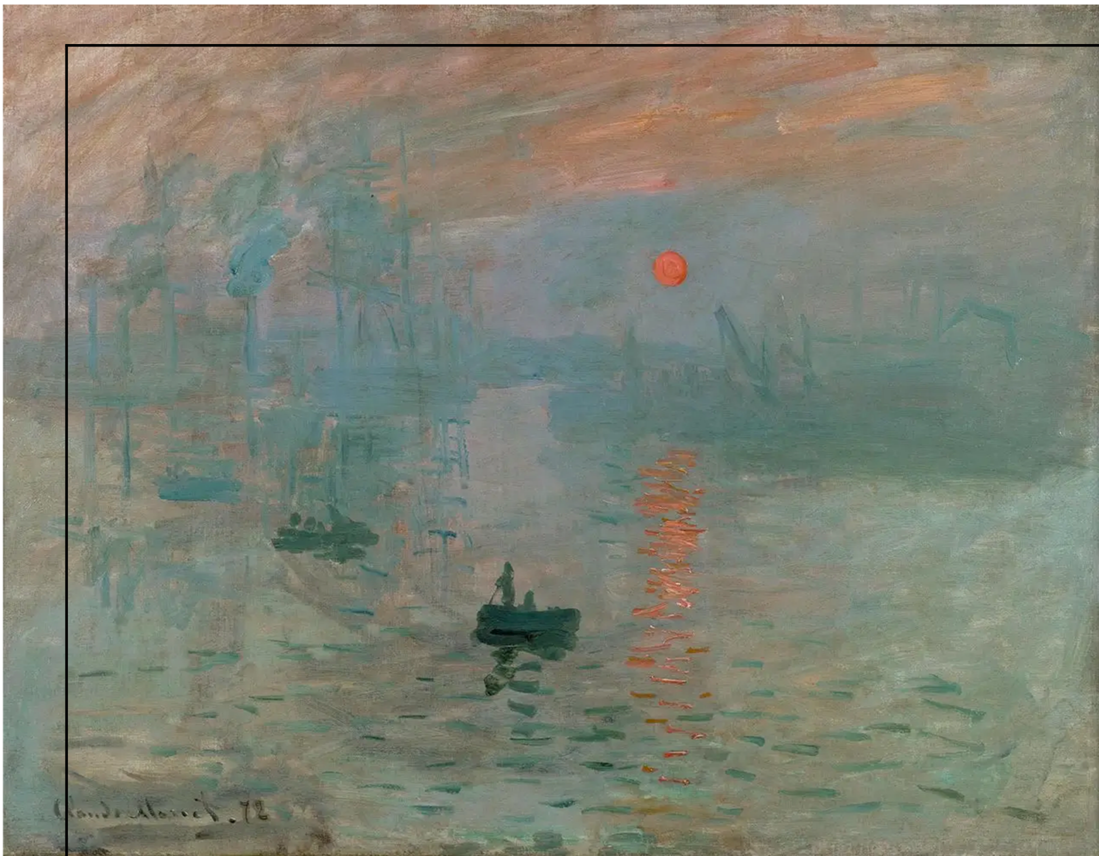
02

General Relativity - PGF5206

Special Relativity

Lucas R. Ximenes
11917239



28
03

Q. 01 Lorentz transformations in terms of rapidity

The rapidity ϕ of a particle moving with velocity u is defined by $\tanh\phi = u$. Prove that the non-trivial equations of the Lorentz transformation can be written as

$$x' = x\cosh\phi - ct\sinh\phi$$
$$ct' = -x\sinh\phi + ct\cosh\phi$$

and use this to verify the following group property (transitivity) of the Lorentz transformation: The resultant of two Lorentz transformations with parameters v_1 and v_2, respectively, is another Lorentz transformation with parameter given by $v = \dfrac{v_1 + v_2}{1 + \frac{v_1 v_2}{c^2}}$.

Q. 02 Some particle properties

A particle of rest mass m and 4-momentum p is examined by an observer with 4-velocity u. Show that (with $c = 1$ and $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$)

- The energy she measures is $E = -p \cdot u$;
- The rest mass she attributes to the particle is $m^2 = -p \cdot p$;
- The momentum she measures has magnitude $|\mathbf{p}|^2 = (p \cdot u)^2 + p \cdot p$;
- The ordinary velocity she measures has magnitude

$$|\mathbf{v}| = 1 + \frac{p \cdot p}{(p \cdot u)^2}$$

- a 4-vector v, whose components in the observer's Lorentz frame are

$$v^0 = 0, \quad v^j - \left.\frac{\mathrm{d}x^j}{\mathrm{d}t}\right|_{\text{particle}} = \text{ordinary velocity}$$

  is given by $v = -u - \dfrac{p}{p \cdot u}$.

Q. 03 The little group of p

Suppose in some inertial frame S a photon has 4-momentum components (with $c = 1$ and $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$)

$$p^0 = p^x = 0E, \quad p^y = p^z = E$$

There is a special class of Lorentz transformations, called the "little group of p", which leaves the components of p unchanged, for instance a pure rotation through an angle α in the $y - z$ plane,

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\alpha & -\sin\alpha \\ 0 & 0 & \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} E \\ E \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} E \\ E \\ 0 \\ 0 \end{pmatrix}$$

is such a transformation. Find a sequence of pure boosts and pure rotations whose product is *not* a pure rotation in the $y - z$ plane, but *is* in the little group of p.

Q. 04 Rotation operators and boosts

Let J_x, J_y and J_z be infinitesimal rotation operators defined so that $1 + iJ_j\theta/2$ is a rotation by a small angle θ around the j-axis. Let K_x, K_y and K_z be infinitesimal boost operators so thaat $1 + iK_jv/2$ is a boost by a small velocity v in the j-direction. Show that the following relationships, and all their cyclic permutations, are true:

$$[J_x, J_y] = 2iJ_z$$
$$[J_x, K_y] = 2iK_z$$
$$[K_x, K_y] = -2iJ_z$$

Find a representation of the Lorentz group in terms of Pauli spin matrices σ_x, σ_y and σ_z and the unit matrix.

Q. 05 Pure boosts

What is the least number of pure boosts which generate an arbitrary Lorentz transformation?

Q. 06 Photon vs. a stationary electron

A photon of wavelength λ hits a stationary electron (mass m_e) and comes off with wavelength λ' at an angle θ. Derive the expression:

$$\lambda' - \lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

Q. 07 Free electron property

Show that it is impossible for an isolated free electron to emit a photon.

Q. 08 $A \to B + C$ process

Consider the reaction $A \to B + C$ (with particle masses m_A, m_B and m_C). For simplicity consider units such that $c = 1$.

- If A is at rest in the lab frame, show that in the lab frame particle B has energy

$$E_B = \frac{m_A^2 + m_B^2 - m_C^2}{2m_A}$$

- An atom of mass M at rest decays to a state of rest energy $M - \delta$ by emitting a photon of energy $h\nu$. Show that $h\nu < \delta$;
- If A decays while moving in the lab frame, find the relation between the angle at which B comes off, and the energies of A and B.

Q. 09 Consistency of special relativity

A new force field $F^\mu(x^\nu)$ is discovered which induces a 4-acceleration $a^\mu = \frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = m^{-1}F^\mu(x^\nu)$ on a particle of mass m at position x^ν. Notice that F^μ dos not depend on u^ν. Show that this force is not consistent with special relativity.

Q. 10 Spacelike and timelike intervals

If two events are separated by a spaceelike interval, show that

- There exists a Lorentz fram in which they are simultaneeus;
- In no Lorentz frame do they occur at the same point.

On the other hand, if two events are separeted by a timelike interval, show that

- There exists a frame in which they happen at the same point;
- In no Lorentz frame are they simultaneous.

Q. 11 Tests with tensors

You are given a tensor $k^{\alpha\beta}$. How can you test whether it is a direct product of two vectors $k^{\alpha\beta} = A^{\alpha}B^{\beta}$? Can you express the test in coordinate-free language?

Q. 12 Second-rank tensors

Prove that the general second-rank tensor in n-dimensions cannot be represented as a simple direct product of two vectors, but can be expressed as a sum over many such products.

Q. 13 A two index object

A two index object $X^{\mu\nu}$ is defined by the "direct sum" of two vectors $X^{\mu\nu} = A^{\mu} + B^{\nu}$. Is $X^{\mu\nu}$ a tensor? Is there a transformation law to take X to a new coordinate system, i.e., to obtain $X^{\hat{\mu}\hat{\nu}}$ from $X^{\mu\nu}$?

Q. 14 Antisymmetric second-rank tensors

Show that a sscond-rank tensor $F_{\mu\nu}$ which is antisymmetric in one coordinate frame is antisymmetric in all frames. Show that the contravariant components are also antisymmetric. Show that symmetry is also coordinate invariant.

Q. 15 High-rank tensors

In a n-dimensional metric space, how many independent components are there for a r-rank tensor $T^{\alpha\beta\cdots}$ with no symmetries? If the tensor is symmetric on s of its indices, how many independent components are there? If the tensor is antisymmetric on a of its indices, how many independent components are there?

Q. 16 Antisymmetric tensors

Show that if the antisymmetric tensor $p^{\alpha\beta}$ is a bivector, $p^{\alpha\beta} = A^{[\alpha}B^{\beta]}$, then

$$p^{\alpha\beta}p^{\gamma\delta} + p^{\alpha\gamma}p^{\delta\beta} + p^{\alpha\delta}p^{\beta\gamma} = 0$$

Recall that $V^{[a_1 a_2]} = \dfrac{1}{2}(V^{a_1 a_2} - V^{a_2 a_1})$.